SPS-24G-01

SPS-24G-01 — Electrical Safety Review

SAMAY ELECTRONICS · Rev A0 · 2026-10-25

> This is an engineering safety review of a mains-connected design, not a certification. It identifies what the layout achieves, what must still be verified, and the standards that apply. **No unit may be energised on mains, sold, or installed until a qualified engineer has reviewed it and the product has passed formal type testing.** Working on 230 VAC is lethal — only trained personnel, with the board de-energised and discharged, and an isolation transformer on the bench.

---

1. Applicable standards (India + international)

Area	Standard
Household automatic/electronic switches	**IEC 60669-1** / **60669-2-1**, BIS **IS 60669**
Electronic switches for appliances	IEC 61058-1
Insulation coordination (creepage/clearance)	**IEC 60664-1**
Safety of SELV / isolation	IEC 61140, IEC 62368-1 (reference)
Surge / EMC	IEC 61000-4-5 (surge), CISPR 14 / IEC 55014 (emissions), 24 GHz ISM radar: FCC Part 15 / CE EN 300 440 (met by the HLK-LD2410B module)
Safety-critical components	Fuse IEC 60127, MOV IEC 61051, X-cap IEC 60384-14 (X2/Y), opto UL1577 / IEC 60747-5-5

---

2. Isolation architecture (the core safety feature)

Two galvanically separate domains:

- **PRIMARY (hazardous live):** J1 terminal, F1, MOV1, CX1, RB1, NTC1, the HLK-PM01 **input**, the MOC3063 **output** (pins 4/6), Q1 triac, RG1, snubber RS1/CS1.
- **SECONDARY / SELV (safe to touch):** +5 V rail, ATtiny13A, LD2410B radar, button, LED, trimmer, ISP and config headers, the HLK-PM01 **output**, the MOC3063 **input** (pins 1/2).

They are placed on **opposite halves** of the board (primary = top strip, SELV = bottom strip) separated by a **6 mm no-copper corridor** that runs the full width, reinforced by a **routed milling slot** (two segments, Edge.Cuts). No copper net crosses the corridor. The only two components bridging it are:

Device	Isolation rating	Role
HLK-PM01	4 kVAC reinforced (AC-DC)	Powers the SELV side from mains
MOC3063	7.5 kVrms (opto)	Carries the MCU's ON/OFF command to the triac gate

Both are the **reinforced/double-insulation barrier** between live mains and the user-touchable SELV electronics. This is the primary protection against electric shock.

Enforced separation (from the layout script `isolation_check`)

- Minimum **primary-pad ↔ SELV-pad centre distance: ≥ 7.0 mm** (target ≥ 6 mm creepage edge-to-edge → met, because the 6 mm corridor keepout forbids any copper between the domains).
- **No primary-net copper exists below the HV/SELV boundary**, and no SELV copper above it (verified automatically every build; the check fails the build otherwise).
- Corridor keep-out is a DRC rule area (no tracks / vias / pours), so the autorouter cannot bridge it.

Requirement to confirm at test: HIPOT **3 kVAC, 1 min, primary-to-secondary**, leakage < 5 mA, no breakdown. The HLK-PM01 and MOC3063 ratings support this; the PCB barrier (≥ 6 mm + slot) supports ≥ 6 mm creepage which exceeds the IEC 60664-1 reinforced requirement for 250 V working (≈ 5.0 mm, PD2, group IIIb).

---

3. Creepage & clearance analysis

3.1 Primary ↔ Secondary (reinforced) — PASS by design

Metric	Required (250 V, reinforced, PD2)	This board
Clearance	≈ 3.0 mm	≥ 6 mm (corridor) ✔
Creepage	≈ 5.0 mm	≥ 6 mm (corridor) + slot ✔

3.2 Primary live-to-live (L / N / LAMP — functional/basic) — REVIEW + COATING

For 250 V working, basic insulation, PD2, material group IIIb: clearance ≈ 1.5 mm, **creepage ≈ 2.5 mm**.

The limiting spots are **component pin pitches**, which are fixed by the packages and *cannot* be widened:

Location	Spacing	Note
BTA16 TO-220 MT1–MT2–Gate	2.54 mm pitch → ~1.0 mm copper gap	Inherent to TO-220. Gate–MT1 carry the gate trigger; MT1–MT2 see mains only when OFF.

MOC3063 output pins 4–6	2.54 mm pitch → ~1.0 mm gap	Inherent to DIP.
HLK-PM01 AC pins 1–2	5.0 mm → ~3 mm gap	OK
J1 terminal L–N–LAMP	5.08 mm pitch	Terminal body provides creepage; OK
Routed L vs N/LAMP traces	≥ 0.8 mm (DRC rule)	**Below 2.5 mm in places**

Mitigations (mandatory for production):

1. Apply **conformal coating** (acrylic/urethane) to the primary side — IEC 60664-3 allows reduced spacing under type-2 coating.
2. Manually widen routed L↔N / L↔LAMP spacing to ≥ 2.5 mm where board area allows (the current board is autorouted at 0.8 mm functional clearance; a HV hand-route pass is a required follow-up).
3. Optionally add **slots between the BTA16 pins** and under the opto to raise pin-pin creepage.
4. Keep pollution degree 2 (sealed bathroom enclosure, IP44+).

---

4. Component-level protection review

Part	Function	Verify
F1 500 mA T, 250 V, TR5	Fault/overcurrent — protects PSU & triac, isolates a shorted board	Breaking capacity ≥ prospective fault current (≥ 1.5 kA for a branch circuit); time-lag so it survives lamp inrush
MOV1 14D471K (S14K300)	Surge clamp L-N, 6 kV/3 kA test	Energy rating for the install; consider a **thermally-fused MOV** for end-of-life safety (MOVs can fail short/overheat)
NTC1 5D-9	Inrush limiter into the PSU cap	In series with the HLK-PM01 only (not the lamp). Hot-restart derating OK for a bathroom
CX1 100 nF **X2** 275 VAC	Differential EMI	Must be an **X2 safety cap**, not a general MLCC
RB1 1 MΩ	Discharges CX1 when unplugged	< 1 s discharge; rated ≥ 350 V working
Q1 BTA16-600B	Load switch	Derate to ~2 A without heatsink; snubber fitted; isolated tab (TO-220 ins.)
RG1/RS1/CS1	Gate + snubber	RS1 ≥ 2 W; CS1 must be an **X1/400 VAC film** cap, not ceramic
PS1 HLK-PM01	Isolated SELV supply	Genuine Hi-Link part; reinforced isolation is the main barrier
U2 MOC3063	Isolated gate drive	Zero-cross type; per ON AN-3004 verify peak opto current < 1 A ITSM (180 Ω gate R is acceptable for a zero-cross opto; raise to 360 Ω if substituting a random-phase opto)

---

5. Protective-earth / enclosure

This is a **Class II (double-insulated)** design — there is **no protective earth** on the PCB. That places the full shock-protection burden on the primary↔secondary isolation (Section 2). The enclosure must be all-insulating (FR-ABS/PC), with no exposed metal reachable by the user, and the sensor window must be RF-transparent plastic with no metallised coating.

---

6. Required type tests before release

1. **HIPOT** 3 kVAC 1 min primary↔secondary (and to any accessible metal) — no breakdown, < 5 mA.
2. **Insulation resistance** > 2 MΩ at 500 VDC primary↔secondary.
3. **Creepage/clearance** physical measurement vs IEC 60664-1, with coating accounted per 60664-3.
4. **Surge** IEC 61000-4-5: 2 kV L-N, 4 kV L-PE (if applicable), 5 strikes each polarity — survives.
5. **Inrush & fuse coordination**: confirm F1 does not nuisance-trip on cold lamp inrush; confirm a bolted output fault opens F1 without the triac/PCB catching fire.
6. **Thermal**: full-load triac + PSU temperature rise under worst-case ambient (sealed box, 40 °C).
7. **EMC**: conducted/radiated emissions (CISPR 14); the 24 GHz radar uses its module's existing FCC/CE certification (do not modify the antenna).
8. **Ball-pressure / glow-wire** on the enclosure and PCB per IS 60669.

---

7. Open items / known limitations (this revision)

- ☐ HV primary routed-trace creepage auto-routed at 0.8 mm — **needs a manual HV routing pass to ≥ 2.5 mm and/or conformal coating** (Section 3.2).
- ☐ MOV is not thermally fused — consider TMOV or series thermal fuse.
- ☐ No in-rush/short-circuit test performed (simulation/paper only).
- ☐ Firmware fail-safe behaviour (radar loss → lamp OFF) must be validated on hardware.
- ☐ Minimum-load behaviour with real LED lamps must be bench-verified (triac latching).
- ☐ BIS/IEC certification not performed.

Bottom line: the isolation barrier (the thing that keeps users safe) is designed correctly and verifiably — ≥ 6 mm creepage, reinforced-rated bridging parts, slot, no copper crossing. The primary-side *internal* spacing and the formal test campaign are the outstanding work before this can be a product.